WATERWATCH · MARKET AND VIABILITY

WaterWatch market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: segments, cited benchmarks, unit cost from the BOM and the open route

WaterWatch market and viability

WaterWatch is for small scheme operators who chlorinate rural piped water but can only check the residual when a technician visits; the instruments that watch chlorine continuously cost ten times its parts cost and need mains power. An open, solar design that a regional workshop can build and an operator can maintain fills that gap, if the sensor holds its calibration (WWT-RSK-001 P1). Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Rural piped schemes with public tapstands	Scheme operators and maintenance service providers, East Africa first (WWT-DEC-001)	Hach CL17sc colorimetric chlorine analyser with installation kit, without reagents: USD 3,454 (USABlueBook, checked 2026-10-25); it needs mains power and reagents	325 (parts, option A, `bom/bom.csv`, indicative)	Continuous free chlorine and turbidity at the tap, off grid, with SMS alerts, for about a tenth of the analyser's price
NGO and emergency chlorination programmes	WASH programmes, cholera and flood response	Manual DPD testing by a technician; a handheld DPD photometer, Hanna HI97711: USD 379.99 to 599.99 (Hanna Instruments, checked 2026-10-25)	325	Readings every hour between visits, not once a visit; the photometer stays the calibration reference
Research and teaching	Universities, WASH research groups, makerspaces	Amperometric free chlorine probe alone, Sensorex FCL: about USD 1,900 (cited in WWT-PRB-001, 2026-09-25); ORP kit, Atlas Scientific: about USD 175 (cited in WWT-PRC-001, 2026-09-25); neither page could be re-opened on 2026-10-25, so both are to re-check	325	An inspectable, reproducible system to test low-cost sensors against lab methods
Option B (reagent colorimetric analyser)	As the first segment	As above	About 380 to 450 (estimate from the option A BOM less the flow cell and sensors, plus two pumps, a stirrer, a larger cabinet and reagents; to assess)	Reference-method chemistry, at the cost of reagents to resupply
Option C (probes in the riser)	As the first segment	As above	About 300 to 380 (estimate: no pole or flow cell, plus pH and ORP electrodes and a manifold; to assess)	No water used, but a cut riser and a loose chlorine estimate

Running costs per unit, outside the parts cost: airtime of about USD 1 to 3 a month (estimate, WWT-PRS-001 assumptions), DPD reagent for the caretaker's comparator, and, for option A, a chlorine electrode refresh at a rate to be found at TRL 4.

Open route

- **Who builds it.** A regional electronics or water workshop builds units from the published files: two cut aluminium plates, a machined or printed flow cell, printed holders, bought modules and plumbing. The scheme operator installs and maintains them with the caretakers.
- **How it spreads.** Through the operator's own schemes first, then through WASH networks, universities that test low-cost sensors, and open hardware communities. Build reports and design suggestions come back through the repository's issue templates.
- **What keeps it open.** Hardware under CERN-OHL-S-2.0, software under MIT (PR-015); every change is published; no cloud service is required to receive the data (PR-006), so an operator is never locked to a vendor.
- **What it does not do.** It is a monitoring aid. It does not replace accredited laboratory testing and never declares water safe (PR-013, WWT-REG-001 R1).

Sources

- USABlueBook, "Hach CL17sc Chlorine Analyzer with Pressure Regulator Installation Kit (no reagents included), 8572400," https://usabluebook.com/hach-cl17sc-chlorine-analyzer-with-pressure-regula-53224, accessed 2026-10-25.
- Hanna Instruments, "Free and Total Chlorine Portable Photometer, HI97711," https://hannainst.com/free-and-total-chlorine-portable-photometer-hi97711.html, accessed 2026-10-25.
- Sensorex, "FCL Amperometric Free Chlorine Sensor," https://sensorex.com/product/fcl-amperometric-free-chlorine-sensor/, as cited in WWT-PRB-001 (2026-09-25); not re-opened on 2026-10-25, to re-check.
- Atlas Scientific, "EZO Complete ORP Kit," https://atlas-scientific.com/kits/ezo-complete-orp-kit/, as cited in WWT-PRC-001 (2026-09-25); not re-opened on 2026-10-25, to re-check.
- Unit cost: `bom/bom.csv` and WWT-CAL-001 K1 (indicative prices, not quotes).